

SOFTWARE REQUIREMENTS SPECIFICATION

MotorX

**Second-Hand Vehicle Marketplace with Intelligent Search
and Automated Inventory Processing**

Version 1.0

02/Aug/2026

Prepared by: Group 23

Revision History

Date	Version	Description	Author
02/Aug/2026	0.1	Initial SRS structure and requirements	Group 23
04/Aug/2026	0.5	Completed functional and non-functional requirements, user stories, use cases, and supporting information	Group 23
05/Aug/2026	1.0	Finalized Software Requirements Specification	Group 23

Member Name	Index Number	Key Contributions & Responsibilities
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1. Introduction

This Software Requirements Specification (SRS) defines the functional and non-functional requirements of the **MotorX Second-Hand Vehicle Marketplace and Inventory Management System**. The system provides a digital platform through which vehicle dealers can manage and upload their vehicle inventory, while buyers can discover and search for available vehicles using conventional filters and intelligent natural-language search.

The system is implemented as a web-based application consisting of a React + TypeScript frontend, an Express.js modular monolithic backend, a dedicated asynchronous ETL worker, and supporting infrastructure for authentication, object storage, background job processing, and intelligent search.

The SRS specifies the external behaviour, functional capabilities, quality requirements, constraints, interfaces, database requirements, and other factors necessary to develop and verify the system.

1.1 Purpose

This SRS fully describes the external behaviour of the application and defines the requirements required for software design, implementation, testing, and acceptance. It is intended for software developers, software architects, testers, project supervisors, stakeholders, and future maintainers.

1.2 Scope

The system is a web-based vehicle marketplace and dealer inventory management platform designed to support the management, processing, and discovery of second-hand vehicle listings.

The system provides two major functional areas:

Dealer and Inventory Management

Registered users may apply for dealer accounts. Dealer account requests shall be reviewed and approved by authorized administrators before dealer-specific functionality becomes available.

Authorized dealers can:

- Register and authenticate.
- Manage dealer information.
- Upload and manage multiple images for vehicle listings.
- Upload vehicle inventory in bulk using CSV files.
- Monitor upload and processing status.
- Review rejected inventory records.
- Correct invalid data and re-upload records.
- Receive processing notifications.

The system processes uploaded inventory through an asynchronous ETL pipeline that performs:

- Data extraction.
- Normalization.
- Validation.

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- Standardization.
- Duplicate detection.
- Categorization.
- Data enrichment.
- Embedding generation for semantic search.
- Storage of valid records.

Buyer Marketplace and Search

Buyers can:

- Browse available vehicles.
- Search using structured filters.
- Search using natural-language queries such as "affordable automatic SUV under 2 million."
- Handle common spelling errors through fuzzy matching.
- Find semantically relevant listings.
- Sort and paginate search results.
- View detailed vehicle information.
- View dealer information associated with a listing.

The system uses a **hybrid intelligent search approach** combining:

1. Structured filtering.
2. Fuzzy matching.
3. Semantic similarity.
4. Relevance ranking.

The system also provides administrative capabilities for managing users, dealers, listings, inventory processing, and system activities.

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
API	Application Programming Interface
AWS	Amazon Web Services
CSV	Comma-Separated Values
ETL	Extract, Transform, Load
ETL Worker	Background process responsible for processing uploaded inventory
Fuzzy Search	Approximate matching that supports common spelling variations
Hybrid Search	Search combining structured, fuzzy, and semantic search

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Firestore Authentication	A managed authentication service used to authenticate users and manage user identities. After successful authentication, Firestore issues an ID token in JWT (JSON Web Token) format. The client includes this token in authenticated API requests, and the backend verifies the token before granting access to protected resources.
JWT (JSON Web Token)	A signed token format used to securely transmit authentication and identity information. In MotorX, Firestore Authentication issues Firestore ID tokens in JWT format, which are verified by the backend before protected resources are accessed.
RBAC	Role-Based Access Control
REST	Representational State Transfer
S3	Amazon Simple Storage Service
SRS	Software Requirements Specification
UI	User Interface
Vector Embedding	Numerical representation used for semantic similarity comparison
Semantic Search	Search based on meaning or contextual similarity
Dealer	Registered business user managing vehicle inventory
Buyer	User who searches for and views vehicle listings
Listing	Vehicle record displayed in the marketplace
Upload Job	Tracked inventory file processing operation
Rejected Record	Inventory row rejected because of validation or processing errors

1.4 References

1. IEEE Computer Society, IEEE Recommended Practice for Software Requirements Specifications, IEEE Std 830-1998.
2. ISO/IEC/IEEE 29148, Systems and Software Engineering — Life Cycle Processes — Requirements Engineering.
3. Project Feasibility Study and Proposal Documents.
4. Official documentation for the selected software technologies.

1.5 Overview

This SRS is organized into six major sections.

Section 1 – Introduction: Describes the purpose and scope of the system, defines terminology, identifies references, and provides an overview of existing systems and technologies.

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Section 2 – Overall Description: Provides a high-level description of the product, including its perspective, major functions, user classes, operating environment, constraints, assumptions, dependencies, and requirements subsets.

Section 3 – Specific Requirements: Defines the detailed functional and non-functional requirements, usability requirements, reliability requirements, performance and security requirements, supportability requirements, design constraints, documentation requirements, external services, interfaces, database requirements, legal requirements, and applicable standards.

Section 4 – User Stories: Restates key system requirements as user stories and acceptance criteria to support development planning, requirements validation, and test-case design.

Section 5 – Use Cases: Describes representative end-to-end workflows involving the primary system actors.

Section 6 – Supporting Information: Contains high-level system diagrams, ETL and intelligent search process descriptions, requirements traceability information, and the system glossary.

The document intentionally does not use formal use-case diagrams. Instead, representative system workflows are described through the detailed use cases in Section 5, while the complete functional behavior of the system is specified in Section 3.1.

1.6 Overview of Existing Systems and Technology

Existing vehicle marketplaces such as Riyasewana and ikman.lk primarily focus on consumer-facing vehicle discovery and individual listings, while enterprise-oriented dealer platforms such as AutoTrader provide more advanced inventory management and syndication capabilities. MotorX aims to combine automated inventory processing, intelligent search, and cloud-portable deployment within a platform targeted at small and mid-sized dealerships.

The planned technology stack includes React with TypeScript for the frontend, Node.js with Express.js for the modular monolithic backend, MongoDB Atlas for persistent data storage, Firebase Authentication for user identity management, Redis with BullMQ for background job processing, MinIO for local object storage during development, AWS S3 for production object storage, Docker Compose for containerized development and deployment, and semantic embedding models for intelligent search. The complete technology stack is described in Section 3.8 (Third-Party Components and External Services).

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2. Overall Description

2.1 Product Perspective

The system is a web-based vehicle marketplace and inventory management platform.

The system consists of:

- A web-based frontend.
- A modular monolithic backend.
- A dedicated asynchronous ETL worker.
- A database.
- Object storage for uploaded files.
- A background job queue.
- A hybrid intelligent search mechanism.

The frontend communicates with the backend through RESTful APIs over HTTPS. User authentication is provided by Firebase Authentication, while the backend is responsible for verifying Firebase ID tokens and enforcing application-specific authorization and business rules. Long-running inventory processing is delegated to a dedicated ETL worker through a background job queue, allowing normal API requests to remain responsive.

The primary application uses a modular monolithic architecture. Business capabilities are separated into logical modules while sharing a single deployable backend. Large inventory processing is separated into an asynchronous worker so that long-running ETL tasks do not block normal API requests.

The backend is logically organized into modules including User & Authentication, Dealer Management, Marketplace, Inventory, Search, Notifications, Administration, and Shared Infrastructure components. The ETL worker shares common contracts and data models with the backend while operating as a separate deployable service.

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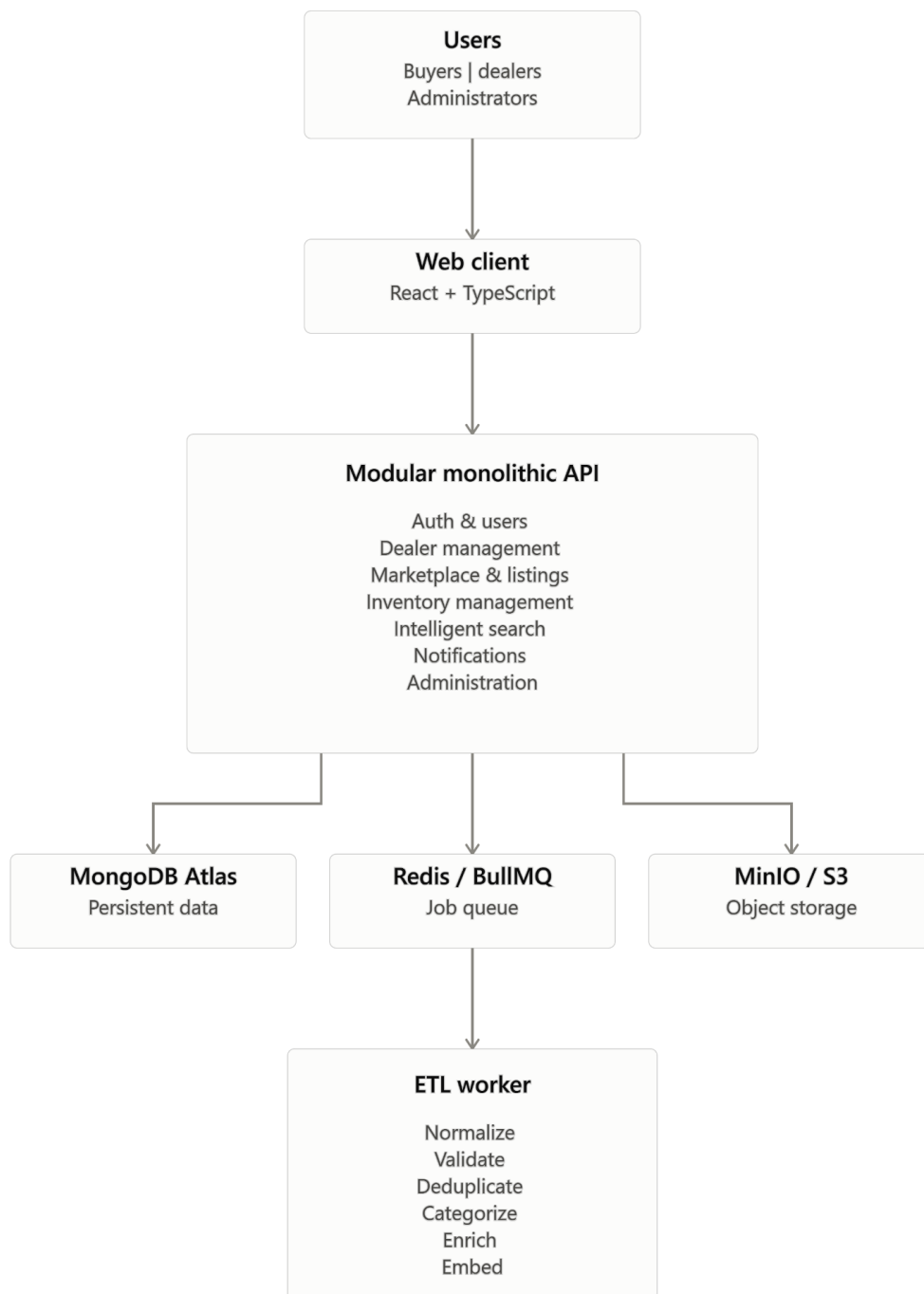


Figure 1: High-Level System Context

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2.2 Product Functions

User and Access Management

- User registration.
- Authentication.
- Firebase ID token verification by the backend
- Role-based authorization.
- Profile management.
- Session management.

Dealer Management

- Dealer registration.
 - Dealer account request submission.
 - Dealer account approval and activation.
- Dealer profile management.
- Dealer inventory management.
- Dealer listing management.

Marketplace

- Vehicle listing creation.
- Vehicle listing updates.
- Listing availability management.
 - Vehicle image management.
- Vehicle detail viewing.
- Dealer information viewing.

Inventory ETL

- CSV upload.
- File storage.
- Upload job creation.
- Background processing.
- Data normalization.
- Validation.
- Duplicate detection.
- Categorization.
- Data enrichment.
- Semantic embedding generation.
- Rejected-row reporting.
- Processing status tracking.

Search

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- Structured search.
- Filter-based search.
- Natural-language query interpretation.
- Structured constraint extraction.
- Fuzzy matching for spelling variations.
- Semantic similarity search.
- Hybrid relevance ranking.
- Sorting.
- Pagination.

Notifications

- Upload completion notifications.
- Upload failure notifications.
- Processing summaries.
- System notifications.
- In-application user notifications.
- Email notifications for significant system events.

Administration

- User management.
- Dealer management.
- Listing management.
- Upload monitoring.
- System activity monitoring.
- System health monitoring.
- Audit information.

2.3 User Classes and Characteristics

User Class	Characteristics and Main Activities
Buyer	Searches vehicles, applies filters, uses intelligent search, views listing details and dealer information.
Dealer	Requests dealer account approval, manages profile, creates listings, uploads CSV inventory, monitors jobs, and reviews rejected records.
Administrator	Manages users and dealers, monitors listings and processing activities, and reviews system activities.

2.4 Operating Environment

The system shall operate as a web-based application using modern web browsers. Production deployment shall use containerized application components and managed cloud services where applicable.

- Client: modern versions of Chrome, Edge, Firefox, or Safari.

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- Client requirements: JavaScript enabled and reliable internet connectivity.
- Server: containerized backend and ETL worker.
- Authentication: Firebase Authentication.
- Persistent storage: MongoDB Atlas.
- Queue: Redis with BullMQ.
- Local object storage (development): MinIO.
- Production object storage: AWS S3 or another S3-compatible object storage service.
- Container orchestration (development): Docker Compose.
- Development: Windows, Linux, or macOS with Node.js, Docker, Docker Compose, MongoDB, Redis, and optional local object storage.

2.5 Constraints

The following constraints influence system development

1. The primary application shall use a modular monolithic backend.
2. Long-running ETL operations shall be separated from normal API processing.
3. Inventory processing shall use asynchronous background jobs.
4. The database shall use MongoDB-compatible storage.
5. Uploaded files shall be stored separately from application metadata.
6. The system shall support containerized development and deployment.
7. The architecture shall remain sufficiently simple for implementation by a small development team.
8. The system shall avoid unnecessary distributed microservice infrastructure.
9. Search shall combine deterministic filtering with intelligent search capabilities.
10. Security-sensitive operations shall require authentication and authorization.
11. User authentication shall be delegated to Firebase Authentication, while application authorization shall be enforced by the backend.
12. Local development shall use Docker Compose to provide a consistent execution environment.

2.6 Assumptions and Dependencies

The system assumes:

- Users have reliable internet access.
- Dealers provide inventory files using supported CSV formats.
- Uploaded files contain sufficient vehicle information.
- Firebase Authentication remains available for authentication operations.
- Cloud storage remains available in production.
- The database remains accessible to the backend.
- The selected embedding model is available during ETL and search processing.
- The system administrator maintains deployment infrastructure.

The system depends on:

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- MongoDB Atlas.
- Redis/BullMQ.
- Object storage.
- Cloud hosting infrastructure.
- Authentication and identity services provided by Firebase.
- Docker and Docker Compose for the standard development environment.
- Embedding model infrastructure.

2.7 Requirements Subsets

Group	Identifier
User Management	FR-USER
Dealer Management	FR-DEALER
Marketplace	FR-MARKET
Inventory Upload	FR-UPLOAD
ETL Processing	FR-ETL
Search	FR-SEARCH
Notifications	FR-NOTIFY
Administration	FR-ADMIN
Usability	UR
Reliability	RR
Performance and Security	PSR
Supportability	SR
Interfaces	IR
Database	DR

3. Specific Requirements

3.1 Functionality

3.1.1 User Registration and Authentication

FR-USER-01: The system shall allow a new user to register by providing the required registration information.

FR-USER-02: The system shall validate required registration fields before creating an account.

FR-USER-03: The system shall prevent duplicate accounts using an already registered Firebase UID or other system-defined unique identity

FR-USER-04: The system shall authenticate registered users before allowing access to protected functionality.

FR-USER-05: The system shall provide appropriate feedback when authentication fails.

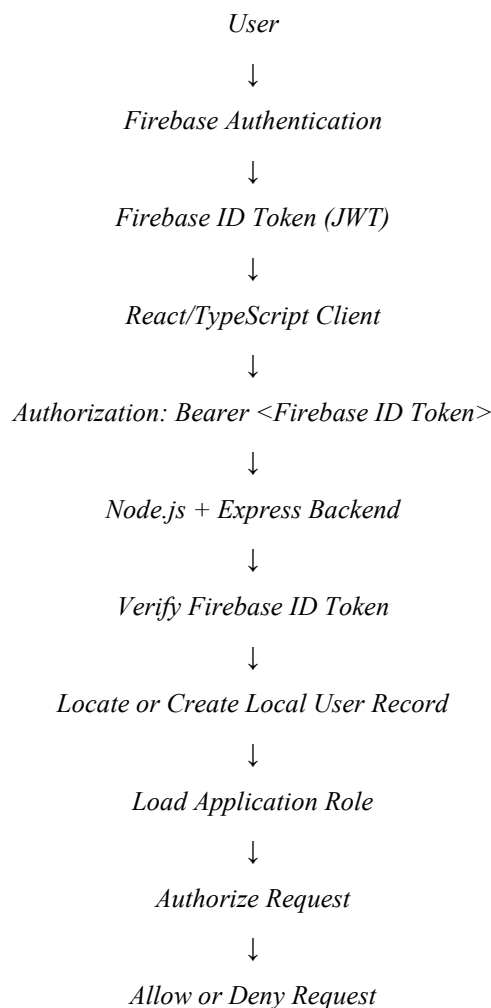
FR-USER-06: The system shall securely maintain authenticated user state using Firebase Authentication mechanisms and verified authentication tokens.

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FR-USER-06A: Following successful Firebase Authentication, the backend shall create a corresponding local user record if one does not already exist and shall associate the authenticated Firebase UID with the application's user profile.

Authentication Architecture:

MotorX shall use Firebase Authentication as the identity and authentication provider. Users shall authenticate through the Firebase Authentication service. Upon successful authentication, Firebase shall issue a Firebase ID token in JWT format to the client. For protected API requests, the client shall include the Firebase ID token in the HTTP Authorization header using the Bearer authentication scheme. The MotorX backend shall verify the Firebase ID token using the Firebase Admin SDK or an equivalent trusted verification mechanism before granting access to protected resources. After successful token verification, the backend shall identify the authenticated user and apply the appropriate role-based authorization rules. Firebase Authentication shall be responsible for authentication and identity management, while the MotorX backend shall remain responsible for application-specific roles and authorization.



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3.1.2 Role-Based Access Control

FR-USER-07: The system shall support at least the following roles: Buyer, Dealer, and Administrator.

FR-USER-08: The system shall restrict functionality according to the authenticated user's role.

FR-USER-09: The system shall prevent buyers from modifying dealer-owned inventory.

FR-USER-10: The system shall restrict administrative functionality to authorized administrators.

FR-USER-11: The system shall verify authorization before performing protected operations.

FR-USER-12: The system shall verify resource ownership before permitting users to modify dealer-owned resources.

3.1.3 Dealer Profile Management

FR-DEALER-01: The system shall allow authorized dealers to create and maintain a dealer profile.

FR-DEALER-02: Dealer profiles shall support business name, contact information, address or location, description, and contact details.

FR-DEALER-03: The system shall allow authorized dealers to update their profile information.

FR-DEALER-04: The system shall display relevant dealer information to buyers when viewing a vehicle listing.

FR-DEALER-05: The system shall provide authorized dealers with a dashboard for monitoring their vehicle inventory and inventory-related activities.

FR-DEALER-06: The system shall display the total number of vehicle listings associated with the authenticated dealer.

FR-DEALER-07: The system shall display inventory upload history and relevant processing statistics for the authenticated dealer.

FR-DEALER-08: The system shall allow authorized dealers to view individual vehicle listings associated with their dealer account.

FR-DEALER-09: The system shall allow a registered user to submit a request to create or activate a dealer account.

FR-DEALER-10: The system shall allow authorized administrators to approve or reject dealer account requests.

FR-DEALER-11: Following approval of a dealer account request, the system shall activate the dealer profile and assign the Dealer application role to the associated user.

FR-DEALER-12: The system shall prevent users without an approved Dealer role from accessing dealer inventory management functionality.

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FR-DEALER-13: The system shall record the approval timestamp and approving administrator for each approved dealer account.

3.1.4 Vehicle Listing Management

FR-MARKET-01: The system shall allow authorized dealers to create vehicle listings.

FR-MARKET-02: A vehicle listing shall support make, model, year, price, mileage, fuel type, transmission, category, location, description, and availability status.

FR-MARKET-03: The system shall validate required listing fields before saving a listing.

FR-MARKET-04: The system shall allow dealers to update their own listings.

FR-MARKET-05: The system shall allow dealers to change listing availability.

FR-MARKET-06: The system shall prevent unauthorized users from modifying another dealer's listings.

FR-MARKET-07: The system shall display available vehicle listings to buyers.

FR-MARKET-07A: Every vehicle listing shall be associated with exactly one approved dealer account.

3.1.5 Vehicle Listing Viewing

FR-MARKET-08: The system shall allow buyers to view detailed information about a selected vehicle.

FR-MARKET-09: The vehicle detail page shall display available vehicle attributes.

FR-MARKET-10: The system shall display relevant dealer information associated with the vehicle.

FR-MARKET-11: The system shall indicate whether a listing is currently available.

FR-MARKET-12: The system shall allow authorized dealers to upload one or more images for each vehicle listing.

FR-MARKET-13: The system shall validate uploaded vehicle image file types and file sizes.

FR-MARKET-14: Vehicle images shall be stored using the configured object storage service while image metadata shall be maintained within the application's database.

FR-MARKET-15: The system shall allow authorized dealers to remove or replace images associated with their own vehicle listings.

FR-MARKET-16: The system shall display available vehicle images to buyers when viewing a vehicle listing.

3.1.6 Inventory File Upload

FR-UPLOAD-01: The system shall allow authorized dealers to upload inventory files in the supported CSV format.

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FR-UPLOAD-02: The system shall validate the uploaded file type before processing.

FR-UPLOAD-03: The system shall reject unsupported file formats.

FR-UPLOAD-04: The system shall store the original uploaded inventory file in the configured object storage service while maintaining upload metadata within the application's database.

FR-UPLOAD-05: The system shall create an upload job record for each accepted inventory upload.

FR-UPLOAD-06: An upload job shall include an identifier, dealer identifier, file information, upload timestamp, processing status, and processing statistics.

FR-UPLOAD-07: The system shall allow dealers to view the status of their upload jobs.

FR-UPLOAD-08: The system shall associate every upload job with the authenticated dealer who initiated the upload.

3.1.7 Asynchronous ETL Processing

FR-ETL-01: The system shall process uploaded inventory files asynchronously.

FR-ETL-02: The system shall not block normal user interactions while an inventory file is being processed.

FR-ETL-03: The system shall place accepted inventory processing jobs into a background job queue.

FR-ETL-04: The ETL Worker shall retrieve queued jobs and process them independently from the main API request.

FR-ETL-05: The ETL Worker shall extract vehicle records from uploaded CSV files.

FR-ETL-06: The ETL process shall process inventory records in manageable batches.

3.1.8 Data Normalization and Validation

FR-ETL-07: The ETL process shall normalize supported vehicle fields into a consistent internal format.

FR-ETL-08: The system shall validate required fields.

FR-ETL-09: The system shall validate data types.

FR-ETL-10: The system shall identify invalid values.

FR-ETL-11: The system shall identify missing mandatory information.

FR-ETL-12: The system shall record validation errors at individual-record level.

The ETL normalization process shall standardize supported vehicle fields into a consistent internal representation. This may include normalizing location names, removing unnecessary whitespace, standardizing capitalization, and converting supported price representations into numeric values. For example, values such as "colombo", "COLOMBO", and "Colombo" may be normalized to "Colombo", while supported representations such as "25 lakh" and "2.5M" may be converted into their corresponding numeric values.

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3.1.9 Partial Processing and Rejected Records

FR-ETL-13: The system shall support partial success during ETL processing.

FR-ETL-14: Valid records shall be imported even when other records in the same file are invalid.

FR-ETL-15: Invalid records shall not be inserted as valid vehicle listings.

FR-ETL-16: The system shall store rejected records separately.

FR-ETL-17: Each rejected record shall contain sufficient information to identify the original record and rejection reason.

FR-ETL-18: The system shall provide processing summaries containing total, successful, rejected, duplicate, and status information.

3.1.10 Duplicate Detection

FR-ETL-19: The system shall detect potential duplicate vehicle records during ETL processing.

FR-ETL-20: Duplicate detection shall use available identifying vehicle attributes.

FR-ETL-21: The system shall prevent confirmed duplicates from creating unintended duplicate listings.

FR-ETL-22: Potential duplicate records shall be recorded in processing results.

3.1.11 Data Categorization and Enrichment

FR-ETL-23: The system shall categorize vehicle data using available vehicle attributes.

FR-ETL-24: The system may derive additional fields from available source data.

FR-ETL-25: Additional source information that does not map directly to the standard schema may be stored as metadata where appropriate.

FR-ETL-26: Derived and enriched fields shall be associated with the correct vehicle record.

3.1.12 Semantic Embedding Generation

FR-ETL-27: The system shall generate semantic vector embeddings for searchable vehicle information.

FR-ETL-28: The embedding generation process shall use a consistent embedding model for inventory processing and search queries.

FR-ETL-29: Embedding generation shall be performed asynchronously as part of inventory processing.

FR-ETL-30: The system shall store generated embeddings in a form that supports semantic similarity search.

3.1.13 Inventory Processing Completion

FR-ETL-31: The system shall update upload job status using at least Pending, Processing, Completed, Completed with Errors, and Failed.

FR-ETL-32: The system shall record processing statistics.

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FR-ETL-33: The system shall notify the dealer when processing is completed or fails.

3.1.14 Structured Vehicle Search

FR-SEARCH-01: The system shall allow buyers to search available vehicle listings.

FR-SEARCH-02: The system shall support filtering by make, model, price range, year range, mileage, fuel type, transmission, category, location, and availability.

FR-SEARCH-03: The system shall support combining multiple filters in a single search.

FR-SEARCH-04: The system shall return only listings satisfying the applied structured filters.

3.1.15 Search Sorting and Pagination

FR-SEARCH-05: The system shall allow sorting by price ascending, price descending, newest listings, vehicle year, and mileage.

FR-SEARCH-06: The system shall paginate large search result sets.

FR-SEARCH-07: The system shall provide sufficient information to navigate between result pages.

3.1.16 Intelligent Natural-Language Search

FR-SEARCH-08: The system shall allow buyers to enter natural-language vehicle search queries.

FR-SEARCH-09: The system shall analyze natural-language queries.

FR-SEARCH-10: The system shall identify recognizable structured constraints from the query.

FR-SEARCH-11: The system shall apply extracted structured constraints to the vehicle dataset.

FR-SEARCH-12: The system shall identify unstructured search intent that cannot be represented through predefined filters.

For example, a natural-language query such as "affordable automatic SUV under 2 million near Colombo" may be interpreted into structured constraints including vehicle category, transmission type, maximum price, and location, while the remaining query intent may be used for fuzzy or semantic relevance matching.

3.1.17 Fuzzy and Semantic Search

FR-SEARCH-13: The system shall support fuzzy matching for recognized typographical errors and spelling variations.

FR-SEARCH-14: The system shall perform semantic similarity matching for relevant unstructured query terms.

FR-SEARCH-15: The system shall use compatible embedding representations for queries and listings.

FR-SEARCH-16: The system shall combine structured filtering and semantic relevance when generating intelligent search results.

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3.1.18 Hybrid Search Ranking

FR-SEARCH-17: The system shall rank search results according to relevance to the user's query.

FR-SEARCH-18: Structured filters shall be applied before semantic ranking where applicable.

FR-SEARCH-19: The system shall return the most relevant matching listings first.

FR-SEARCH-20: The hybrid search mechanism shall combine structured constraint satisfaction, fuzzy matching relevance, and semantic similarity to determine the ranking of search results.

3.1.19 Notifications

FR-NOTIFY-01: The system shall provide in-application notifications to dealers when inventory processing is successfully completed.

FR-NOTIFY-02: The system shall provide in-application notifications to dealers when inventory processing fails.

FR-NOTIFY-03: Processing notifications shall include relevant summary information, including processing status and the number of successful, rejected, and duplicate records where applicable.

FR-NOTIFY-04: The system shall associate notifications with the appropriate authenticated user.

FR-NOTIFY-05: The system shall prevent users from viewing notifications belonging to other users.

FR-NOTIFY-06: The system shall send email notifications for important events such as dealer approval or rejection, inventory processing completion, and inventory processing failure.

FR-NOTIFY-07: Failure to send an email notification shall not prevent the corresponding in-application notification from being created.

3.1.20 Administration

FR-ADMIN-01: The system shall provide administrative functionality to authorized administrators.

FR-ADMIN-02: Administrators shall be able to view and manage user accounts.

FR-ADMIN-03: Administrators shall be able to view dealer accounts and approve or reject dealer account requests.

FR-ADMIN-04: Administrators shall be able to monitor vehicle listings.

FR-ADMIN-05: Administrators shall be able to monitor inventory processing activities.

FR-ADMIN-06: The system shall maintain audit information for significant administrative operations.

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FR-ADMIN-07: The system shall provide authorized administrators with system health information for critical application services.

FR-ADMIN-08: The system shall display the operational status of the backend application, database, background job queue, and ETL worker.

FR-ADMIN-09: The system shall display relevant recent system activities, including recent inventory uploads and processing failures.

FR-ADMIN-10: The system shall provide administrators with system-level statistics, including the total number of vehicle listings and registered dealers.

3.2 Usability

UR-01: The system shall provide a consistent interface across major user functions.

UR-02: The system shall use clear labels for forms, buttons, menus, and navigation elements.

UR-03: The system shall provide meaningful validation messages.

UR-04: The system shall clearly identify mandatory fields.

UR-05: The search interface shall provide clear structured filter controls.

UR-06: The intelligent search interface shall provide a text input area for natural-language queries.

UR-07: Users shall be able to combine natural-language search with structured filters where appropriate.

UR-08: Search results shall clearly display relevant vehicle information.

UR-09: The upload interface shall clearly indicate the supported CSV format.

UR-10: The system shall display upload progress or processing status.

UR-11: The system shall clearly distinguish successful, rejected, duplicate, and failed records.

UR-12: Error messages shall provide enough information for a dealer to correct invalid records.

UR-13: A normal buyer shall be able to perform basic vehicle searches without formal training.

UR-14: A dealer shall be able to learn the basic inventory upload workflow using the provided user documentation within a reasonable training period.

UR-15: Administrative users shall receive system documentation describing administrative functions.

UR-16: The system shall provide a responsive user interface that supports modern desktop, tablet, and mobile web browsers.

UR-17: The system shall adapt major marketplace, search, listing, and dealer dashboard interfaces to different screen sizes.

UR-18: Forms shall provide clear labels and meaningful validation messages to support users in completing required actions.

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3.3 Reliability

RR-01: The production system shall target at least 99% monthly availability, excluding scheduled maintenance.

RR-02: The system shall not create a valid vehicle listing from a record that fails mandatory validation.

RR-03: The system shall preserve original uploaded files for traceability.

RR-04: The system shall maintain relationships between upload jobs, processed records, and rejected records.

RR-05: Failure of an individual inventory record shall not cause the entire valid dataset to be rejected.

RR-06: Failed background jobs shall be recorded.

RR-07: The system shall support retrying recoverable ETL processing failures.

RR-08: The system shall prevent repeated processing from unintentionally creating duplicate listings.

RR-09: The system shall provide meaningful error information without exposing internal system details.

RR-10: Unexpected server errors shall be logged for diagnostic purposes.

3.4 Performance and Security

PSR-01: For normal API operations under expected load, the system shall return successful responses within 2 seconds for at least 95% of requests.

PSR-02: Standard structured searches shall return results within 2 seconds under expected operating conditions.

PSR-03: Intelligent semantic searches should normally return results within 5 seconds, excluding temporary external model-service delays.

PSR-04: The system shall process uploaded inventory files asynchronously.

PSR-05: The system shall be designed to process approximately 5,000 vehicle records within 2 minutes under a defined test environment, subject to enrichment and embedding complexity.

PSR-06: The ETL process shall not block normal API requests.

PSR-07: The system architecture shall support future growth in the number of registered dealers, users, and vehicle listings without requiring major architectural changes.

PSR-08: The system shall require authentication for protected operations.

PSR-09: The system shall enforce role-based authorization.

PSR-10: The system shall use HTTPS for production communications.

PSR-11: The system shall use Firebase Authentication for password-based user authentication and shall not store user passwords in the application's MongoDB database.

PSR-12: Authentication credentials and Firebase service credentials shall be protected using secure configuration and secret management mechanisms.

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PSR-13: The system shall validate and sanitize user-provided input.

PSR-14: Uploaded files shall be validated before processing.

PSR-15: The system shall restrict dealers to their authorized inventory data.

PSR-16: Administrative operations shall require appropriate authorization.

3.5 Supportability

SR-01: The software shall use modular internal organization.

SR-02: The system shall separate presentation, business logic, and data access responsibilities.

SR-03: The ETL worker shall remain logically separated from the main API application.

SR-04: The system shall log significant application errors.

SR-05: The system shall log ETL job status and processing failures.

SR-06: The system shall log significant security and administrative activities.

SR-07: The system shall provide monitoring information for API health, background job status, ETL failures, database connectivity, object storage availability, and application errors.

SR-08: The system shall support containerized deployment.

SR-09: The system shall provide environment-specific configuration.

SR-10: Sensitive configuration shall be provided through environment variables or secure secret management.

3.6 Design Constraints

- **Standards Compliance:** HTTP/HTTPS, REST API conventions, JSON data formats, secure communications, and recognized secure coding practices.
- **Hardware Limitations:** No specialized hardware shall be mandatory. The embedding model shall be selected to operate within practical project infrastructure constraints; GPU hardware shall not be mandatory for normal operation.

3.6.1 Architectural and Technology Constraints

ID	Constraint
DC-01	The primary application shall use a modular monolithic backend with logical modules for Auth & Users, Dealers, Marketplace, Inventory, Search, Notifications, and Administration.
DC-02	ETL processing shall be implemented as a separate asynchronous worker using Redis/BullMQ unless an approved alternative is selected.
DC-03	The technology stack shall use React, TypeScript, Node.js, Express.js, MongoDB, Firebase Authentication, Redis, BullMQ, Docker, and Docker Compose, or compatible approved alternatives where applicable.
DC-04	Original inventory files shall be stored in object storage rather than directly within the primary database. MinIO shall be used for local development and AWS S3 shall be used for production where applicable.

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3.7 Online User Documentation and Help System Requirements

- The system shall provide user documentation for registration, login, vehicle search, intelligent search, dealer account requests, dealer profile management, inventory upload, upload status monitoring, and rejected record handling.
- The system shall provide guidance on the supported CSV inventory format.
- The documentation shall describe common validation errors and their resolution.
- Administrative documentation shall describe administrative operations.

3.8 Third-Party Components and External Services

Component	Purpose
MongoDB Atlas	Managed database
Firebase Authentication	User authentication and identity management
Redis	Background job queue infrastructure
BullMQ	Job processing
AWS S3	Production object storage
MinIO	S3-compatible object storage for local development
Hugging Face Model	Semantic embedding generation
Email Service	Delivery of application email notifications
Docker	Containerization
Docker Compose	Local multi-container development environment
Cloud Hosting	Application deployment

All third-party components shall be used according to their respective licenses and terms. Exact versions and licenses shall be recorded in final implementation documentation.

3.9 Interfaces

The following interfaces define the major interactions required by the system. Detailed UI screenshots are not included because the interfaces are not yet implemented.

3.9.1 User Interfaces

Interface	Required Elements
Login	Firebase Authentication-based login interface, supported authentication credentials, login action, authentication error messages, password recovery where applicable, and registration navigation.
Buyer Marketplace	Search input, intelligent search input, filters, price/year range, fuel, transmission, category, mileage, location, sorting, pagination, listing cards.
Vehicle Details	Vehicle attributes, price, mileage, year, fuel, transmission, category, location, description, dealer information, availability.

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Dealer Dashboard	Dealer profile, listings, add/edit listing, vehicle image management, inventory upload, upload history, processing status, statistics, rejected records, notifications.
Inventory Upload	CSV selection, file validation, upload action, status, progress, result summary.
Administration	User management, dealer management, dealer approval, listing management, upload monitoring, system monitoring, audit information.

3.9.2 Hardware Interfaces

No specialized hardware interface is required. Client devices should have at least a dual-core processor or equivalent, 4 GB RAM, a modern browser, and stable internet connectivity. Production infrastructure shall be sized through performance testing.

3.9.3 Software Interfaces

Interface	Description
Frontend ↔ Backend	RESTful HTTP/HTTPS APIs with JSON request and response formats.
Frontend ↔ Firebase Authentication	User registration, login, authentication state management, and Firebase ID token acquisition.
Backend ↔ Firebase Authentication	Firebase Admin SDK or equivalent mechanism for Firebase ID token verification.
Backend ↔ MongoDB Atlas	Database communication using a supported MongoDB driver or ODM.
Backend ↔ Redis	Background job management through BullMQ.
Backend ↔ Email Service	Sending application email notifications for important system events.
ETL Worker ↔ Redis	Retrieval and completion of queued processing jobs.
Backend/ETL Worker ↔ Object Storage	Upload and retrieval of original inventory files.

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Backend/ETL Worker ↔ Embedding Model	Generation of listing and query embeddings using a consistent compatible model.
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3.9.4 Communications Interfaces

- HTTPS communication between clients and backend.
- REST API requests using JSON.
- Firebase Authentication communication using supported secure Firebase protocols and SDKs.
- Authenticated API requests using Firebase ID tokens in JWT format, transmitted through the HTTP Authorization: Bearer header over HTTPS.
- Asynchronous background job communication through Redis/BullMQ.
- Secure communication between the backend and email service for application email notifications.
- Encrypted communication in production.

Draft interface relationship:

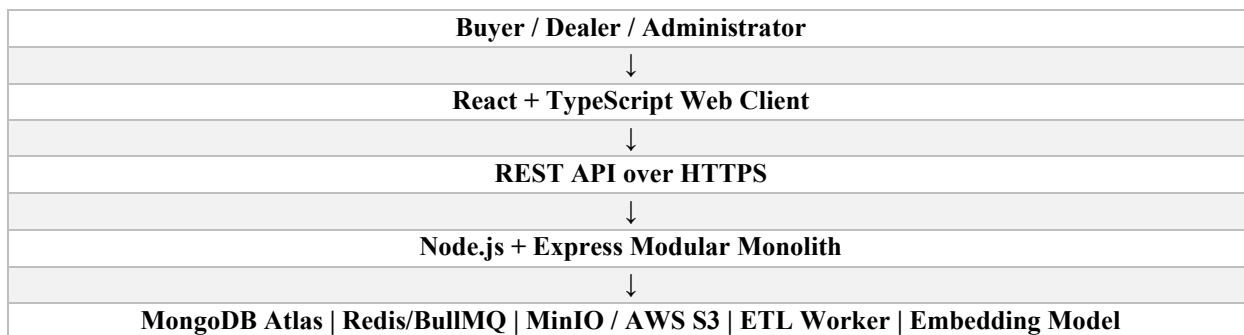


Figure 2. Draft Main Interface Relationship

3.10 Database Requirements

The system shall use MongoDB Atlas as the primary persistent database. Data shall be logically separated into collections representing major business domains.

Collection	Main Data
Users	User ID, Firebase UID, role, account status, profile information, timestamps
Dealers	Dealer ID, associated user ID, business information, contact details, account status
Listings	Listing ID, dealer ID, vehicle attributes, description, condition, location, images, availability, metadata, search data, embedding/vector representation where supported
Upload Jobs	Upload ID, dealer ID, file reference, status, upload timestamp, processing timestamps, record statistics

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Rejected Records	Rejected record ID, upload job ID, original row information, validation errors, rejection reason
Notifications	Notification ID, recipient user ID, notification type, message, read status, email delivery status, timestamp
Audit Logs	Audit log ID, user ID where applicable, operation type, affected resource, timestamp, relevant activity information

The database shall support appropriate indexes for make, model, price, year, fuel type, transmission, category, location, dealer ID, and availability. Search-related indexes shall be designed to support structured filtering and efficient retrieval. The database shall also support efficient retrieval of vehicle images using listing identifiers and notification retrieval using recipient user identifiers.

The database shall also support efficient retrieval of dealer-specific inventory using dealer identifiers and upload-related records using upload job identifiers. Appropriate indexes shall be created for frequently queried fields and relationships, including dealer ID, upload job ID, Firebase UID, availability status, and search attributes.

3.11 Licensing, Legal, Copyright, and Other Notices

The system shall comply with applicable software licensing requirements. Third-party libraries shall be used according to their respective licenses. Proprietary software or models shall not be redistributed without authorization. External embedding models and Firebase services

shall be used according to their model licenses and usage restrictions. User-provided inventory data shall be handled according to applicable privacy and data protection requirements.

3.12 Applicable Standards

1. IEEE 830-1998 — Recommended Practice for Software Requirements Specifications.
2. ISO/IEC/IEEE 29148 — Requirements Engineering.
3. HTTP/HTTPS standards for web communication.
4. REST architectural principles for API design.
5. JSON format for structured API communication.
6. OWASP secure software development recommendations.
7. Applicable software licensing requirements.

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4. User Stories

This section restates key requirements from Section 3 as user stories with acceptance criteria, to support sprint planning and test-case design. Each story traces back to the functional requirements identified in parentheses.

Dealer User Stories

US-01: Bulk Vehicle Upload (FR-UPLOAD-01-07, FR-ETL-01-06, FR-ETL-13-18, FR-NOTIFY-01-05)

As a dealer, I want to upload a CSV file containing multiple vehicles at once, so that I do not have to manually enter each vehicle individually.

Acceptance Criteria:

- I can select a supported CSV file from my computer.
- The system validates the uploaded file before accepting it.
- The system stores the original uploaded file.
- The system creates an upload job for the accepted file.
- The upload job is processed asynchronously.
- I can view the processing status of my upload.
- I receive an in-application notification when inventory processing is completed.
- I can see the number of successfully processed, rejected, and duplicate records.
- If there are invalid records, I can identify the affected rows and their rejection reasons.
- Valid records are imported even when other records in the same file are invalid.

US-02: Monitor Inventory (FR-DEALER-01-08, FR-UPLOAD-07)

As a dealer, I want to see my uploaded vehicles and inventory statistics on a dashboard, so that I can track and manage the vehicles listed under my dealership.

Acceptance Criteria:

- The dashboard shows all vehicle listings associated with my dealer account.
- The dashboard shows my inventory upload history.
- I can see the total number of vehicle listings associated with my dealer account.
- I can view relevant upload processing statistics.
- I can view individual vehicle listing details.
- I can see the processing status of my inventory uploads.
- I cannot view or manage another dealer's private inventory management information.

US-03: Account Security (FR-USER-04-06, PSR-11-12)

As a dealer, I want my account and inventory to be securely protected, so that only authorized users can access my business information.

Acceptance Criteria:

- I must authenticate using the supported Firebase Authentication mechanism.
- The system verifies my Firebase ID token before allowing protected operations.

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- I cannot access another dealer's private inventory management functions.
- I cannot modify another dealer's vehicle listings.
- My application role determines which functions I can access.
- Authentication credentials are managed by Firebase Authentication rather than stored directly in the application's database.

Buyer User Stories

US-04: Browse Vehicles (FR-MARKET-07–11,PSR-02)

As a buyer, I want to browse all available vehicles, so that I can see what's on the market.

Acceptance Criteria:

- I can view a list/grid of vehicles.
- Each vehicle shows key information (price, make, model, mileage).
- I can click on a vehicle to see full details.
- Vehicles load quickly, with no long waiting times.

US-05: Natural-Language Search (FR-SEARCH-08–16)

As a buyer, I want to search for vehicles using plain English, so that I don't have to use complicated filters.

Acceptance Criteria:

- I can type a description of what I want (e.g., "affordable automatic SUV").
- The system returns matching vehicles.
- Results are relevant to my query.
- If there are no results, the system tells me no matches were found.

US-06: Use Filters (FR-SEARCH-01–04)

As a buyer, I want to use filters to narrow down results, so that I can find exactly what I'm looking for.

Acceptance Criteria:

- I can filter by price range.
- I can filter by vehicle type, fuel type, and transmission.
- I can filter by location.
- Filters work together — I can use multiple at the same time.
- Results update as soon as I change filters.

US-07: View Vehicle Details (FR-MARKET-08–11, FR-MARKET-02)

As a buyer, I want to see complete information about a vehicle, so that I can make an informed decision.

Acceptance Criteria:

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- I can see the vehicle specifications, including year, price, mileage, fuel type, transmission, and category.
- I can see the vehicle description and condition when provided.
- I can see vehicle images when they are available.
- I can see the vehicle's location.
- I can see the dealer's relevant contact information.
- The system indicates whether the vehicle is currently available

Administrator User Stories

US-08: Monitor System Health (FR-ADMIN-01–10)

As an admin, I want to see system statistics, so that I can monitor platform usage.

Acceptance Criteria:

- The administrator can view the total number of vehicle listings in the system.
- The administrator can view the total number of registered dealers.
- The administrator can view the operational status of the backend application.
- The administrator can view the database connection status.
- The administrator can view the background job queue status.
- The administrator can view the ETL worker status.
- The administrator can view recent inventory uploads and processing failures.
- Access to system health and administrative information is restricted to authorized administrators.

US-09: Request Dealer Account Activation (FR-DEALER-09, FR-USER-07–08)

As a registered user, I want to request activation of a Dealer account, so that I can list and manage vehicle inventory on the platform.

Acceptance Criteria:

- Can submit a request to activate a Dealer account from my user profile.
- The system records my request as pending until an administrator reviews it.
- Cannot access dealer inventory management functionality until my request is approved.
- Can view the current status of my dealer activation request
- Once approved, I gain access to dealer functionality without needing to re-register.

US-10: Approve or Reject Dealer Account Requests (FR-DEALER-10–12, FR-ADMIN-03, FR-ADMIN-06)

As an administrator, I want to review and approve or reject pending dealer account requests, so that only verified businesses can list vehicle inventory on the platform.

Acceptance Criteria:

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- I can view a list of pending dealer account requests.
- I can approve or reject a pending request.
- A user's Dealer role is activated only after I approve their request.
- A rejected request does not grant dealer access.
- Each approval or rejection decision is recorded in the audit log.
- I cannot approve or reject a request unless I am an authenticated administrator.

5. Use Cases

These use-cases elaborate three representative end-to-end workflows referenced throughout Section 3, each traced to its actor, precondition, main flow, alternative flows, and postcondition.

UC-01: Dealer Uploads Vehicle Inventory

- Actor: Dealer.
- Precondition: Dealer is logged in to the system.

Main Flow:

1. Dealer clicks "Upload Vehicles".
2. Dealer selects a CSV file.
3. The system validates the file.
4. The system uploads the file to object storage.
5. The system creates an upload job.
6. The system places the job in the background queue.
7. The ETL worker processes the file.
8. The system updates the upload status.
9. The system creates an in-application notification and sends an email notification when processing is completed or fails.
10. The system displays the processing summary.
11. The dealer can review rejected records.

Postcondition: Vehicles are now searchable by buyers; the dealer dashboard is updated.

Alternative Flow — Invalid CSV:

If the CSV format is invalid, the system shows an error message, rejects the upload, and instructs the dealer to check the CSV format.

Alternative Flow — Partial Failure:

If some rows are valid and some invalid, valid rows are imported, invalid rows are reported to the dealer, and the dealer can fix the invalid rows and retry.

UC-02: Buyer Searches for a Vehicle

- Actor: Buyer.
- Precondition: The system contains vehicles from dealer uploads.

Main Flow — Natural-Language Search:

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5. Buyer accesses the search page and types a natural-language query (e.g., "affordable automatic diesel SUV under 2 million").
6. The system parses the query to understand intent and searches the database for matching vehicles.
7. The system returns results ranked by relevance, showing year, make, model, price, mileage, location, and fuel type for each.
8. Buyer clicks on a vehicle to see full details.

Postcondition: Buyer can view vehicle details and contact the dealer.

Alternative Flow — Filter-Based Search:

Buyer sets structured filters (e.g., price range, fuel type, transmission, location); the system applies the filters, returns matching vehicles, and the buyer can further sort and open a vehicle's details.

Alternative Flow — No Results:

If the search returns no results, the system shows a "No vehicles found" message and suggests different search terms or filters.

UC-03: Administrator Monitors the System

- Actor: Administrator.
- Precondition: Administrator is logged into the system.

Main Flow:

1. Admin opens the "Dashboard".
2. The system displays total vehicles in the system, total registered dealers, system health status, and upload history.
3. Admin can view details of recent uploads and any errors that occurred.

Postcondition: Admin has visibility into system operations.

6. Supporting Information

The supporting information makes the SRS easier to use. The diagrams and traceability tables in this section are considered supporting requirements information and shall be read together with Section 3.

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Appendix A — High-Level System Context

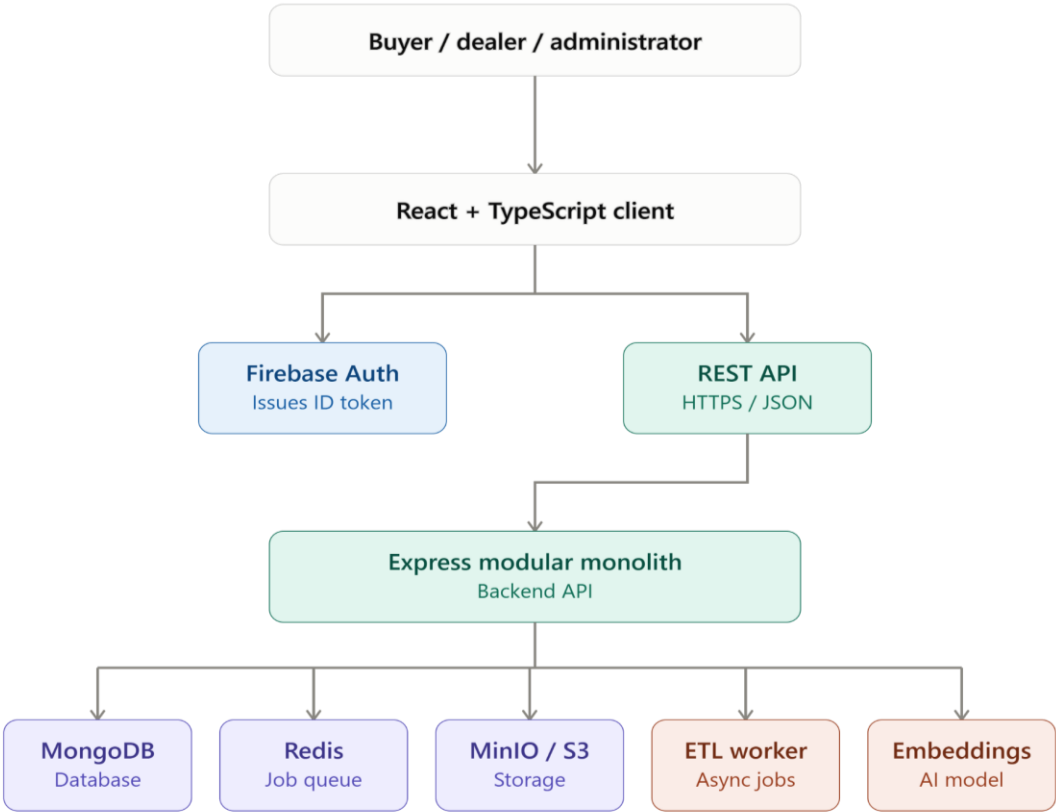


Figure A1. High-Level System Context

Appendix B — ETL Pipeline

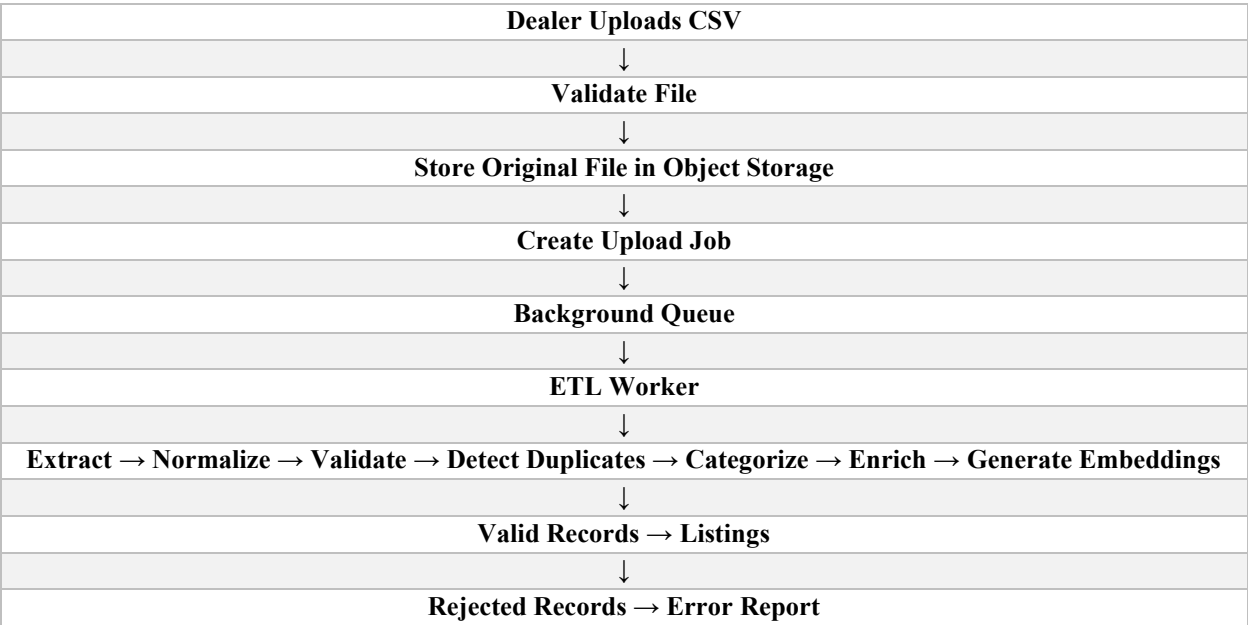


Figure B1. Inventory ETL Pipeline

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Appendix C — Intelligent Search Process

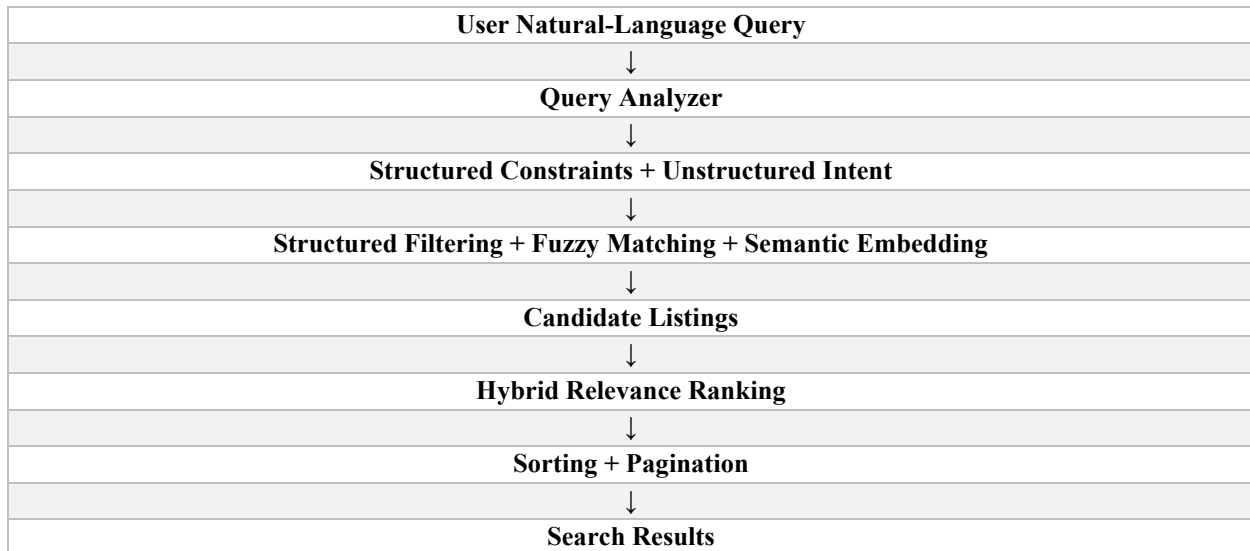


Figure C1. Hybrid Intelligent Search Process

Appendix D — Requirements Traceability

Requirement Area	Architecture Component	Verification Method
Authentication	Firestore Authentication + Auth Integration	Functional Test
Firestore ID Token Verification	Backend Authentication Middleware	Security/Integration Test
RBAC	Authorization Layer	Security Test
Dealer Management	Dealer Module	Functional Test
Listing Management	Marketplace Module	Functional Test
CSV Upload	Inventory Module	Functional Test
ETL	ETL Worker	Integration Test
Validation	ETL Worker	Unit/Integration Test
Duplicate Detection	ETL Worker	Test Dataset
Rejection Reporting	Inventory Module	Functional Test
Structured Search	Search Module	Functional Test
Natural-Language Search	Search Module	Functional/Search Test
Fuzzy Search	Search Module	Search Test
Semantic Search	Search Module	Search Evaluation
Hybrid Ranking	Search Module	Relevance Evaluation
Notifications	Notification Module + Email Service	Functional Test
Administration	Admin Module	Functional/Security Test

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Performance	Complete System	Performance Test
Security	Firebase + Backend	Security Test
Reliability	Backend + Worker	Failure/Recovery Test

Appendix E — Glossary

Term	Definition
Asynchronous Processing	Processing performed independently from the user's immediate request.
Background Job	A task placed in a queue for processing by a worker.
Dealer	A business user responsible for managing vehicle inventory.
ETL	A process that extracts data, transforms it into a standard form, and loads it into the system.
Fuzzy Matching	Approximate text matching that allows spelling variations or typographical errors.
Hybrid Search	A search method combining multiple search strategies.
Intelligent Search	A search mechanism that interprets natural-language queries and retrieves semantically relevant results.
Listing	A vehicle record made available through the marketplace.
Semantic Embedding	A numerical representation of text or data that captures semantic meaning.
Vector Search	Searching for records based on similarity between vector representations.
Upload Job	A tracked process representing the processing of an uploaded inventory file.

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